LAB ASSIGNMENT #05

INTRODUCTION TO FOR LOOPS AND SWITCH CASE

Student Name	Bilal Mirza
Father's Name	Babar Mirza
Section	CSS
Subject	Programming / C++

Lab Objective

To practice for loops and switch-case statements and understand how they are used in C/C++ programs.

Task 1: Print the Table of Any Number

```
#include <iostream>
using namespace std;

int main() {
    int num;
    cout << "Enter a number: ";
    cin >> num;

    for (int i = 1; i <= 10; i++) {
        cout << num << " x " << i << " = " << num * i << endl;
    }
    return 0;
}
```

Task 2: Find the Power of Any Number Using a for Loop

This program calculates base raised to a non-negative integer exponent.

```
#include <iostream>
using namespace std;

int main() {
    int base, exponent;
    long long result = 1;

    cout << "Enter base and non-negative exponent: ";
    cin >> base >> exponent;

    for (int i = 1; i <= exponent; i++) {
        result *= base;
    }

    cout << "Answer = " << result << endl;
    return 0;
}
```

Task 3: Sum of Natural Numbers Using a Loop

```
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
    cout << "Enter a positive integer: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        sum += i;
    }

    cout << "Sum = " << sum << endl;
    return 0;
}
```

Task 4: Sum of Digits of a Number

Example: 358 gives 3 + 5 + 8 = 16.

```
#include <iostream>
using namespace std;

int main() {
    int n, sum = 0;
```

```
    cout << "Enter an integer: ";
    cin >> n;

    if (n < 0) n = -n;
    while (n > 0) {
        sum += n % 10;
        n /= 10;
    }

    cout << "Sum of digits = " << sum << endl;
    return 0;
}
```

Task 5: Generate the Factorial of a Number

```
#include <iostream>
using namespace std;

int main() {
    int n;
    unsigned long long factorial = 1;
    cout << "Enter a number (0 to 20): ";
    cin >> n;

    if (n < 0 || n > 20) {
        cout << "Please enter a number from 0 to 20." << endl;
    } else {
        for (int i = 1; i <= n; i++) {
            factorial *= i;
        }
        cout << "Factorial = " << factorial << endl;
    }
    return 0;
}
```

Task 6: Print the Given Star Pattern

Output: a left-aligned triangle containing 1 to 7 stars.

```
#include <iostream>
using namespace std;

int main() {
    for (int row = 1; row <= 7; row++) {
        for (int col = 1; col <= row; col++) {
            cout << "*";
        }
        cout << endl;
    }
    return 0;
}
```

Task 7: Number of Days in a Month Using switch-case

```
#include <iostream>
using namespace std;

int main() {
    int month;
    cout << "Enter month number (1-12): ";
    cin >> month;

    switch (month) {
        case 1: case 3: case 5: case 7:
        case 8: case 10: case 12:
            cout << "31 days" << endl;
            break;
        case 4: case 6: case 9: case 11:
            cout << "30 days" << endl;
            break;
        case 2:
            cout << "28 days, or 29 in a leap year" << endl;
            break;
        default:
            cout << "Invalid month number" << endl;
    }
    return 0;
}
```

Task 8: Check Vowel or Consonant Using switch-case

```
#include <iostream>
#include <cctype>
using namespace std;

int main() {
    char ch;
    cout << "Enter an alphabet: ";
    cin >> ch;
    ch = tolower(static_cast<unsigned char>(ch));

    switch (ch) {
        case 'a': case 'e': case 'i': case 'o': case 'u':
            cout << "Vowel" << endl;
            break;
        default:
            if (ch >= 'a' && ch <= 'z')
                cout << "Consonant" << endl;
            else
                cout << "Invalid input" << endl;
    }
    return 0;
}
```

Output Questions

1. For `for (5; 2; 2)`, the code is invalid in C/C++ because the first expression is not a valid expression statement in this form (and the loop syntax shown is malformed for a normal counting loop). It produces a compilation error.

2. If input is 6, the loop starts with i = 0 and checks i > 9. This condition is false immediately, so the loop body does not execute. Output: no text.

3. For `for (;;)`, all three expressions are omitted, so it is an infinite loop. It repeatedly prints `10` without stopping (unless the program is interrupted).

4. The last program prints `C++ for loop` and then prints the numbers 0 through 5 without spaces or line breaks.

Conclusion

In this lab, for loops were used for repetition, calculations, patterns, and factorials. Switch-case was used to make choices based on a month number or an alphabet.